

Modern Computer Architecture.

Basics

- Computer architecture is essential for understanding how a computer process instructions and manage hardware.

Components of computer architecture

CPU

- The CPU is the brain of the computer that process instructions and control all operations.

components of CPU:

i) ALU (Arithmetic logic unit)

- ↳ It performs arithmetic and logical operation.

ii) CU = control unit

- ↳ It direct how instruction are executed by managing data flow between components.

iii) Registers

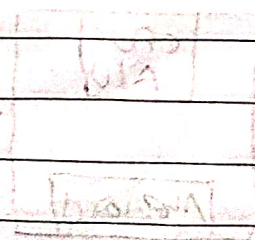
- ↳ It is a small storage location inside the CPU that temporary hold data and instructions for fast access.

→ cache

→ Registers

→ RAM

→ ROM



Input and output system:-

i) Input device

→ Allow user to enter data

ii) Output device

→ Display or produce result.

Memory system:-

- Memory is responsible for storing and retrieving data and instructions

(i) Primary memory:-

RAM → It is a fast and volatile memory used for active process

cache → It is a small high speed memory inside the CPU, that stores frequently used data to improve processing speed.

(ii) Secondary memory:-

- It is a permanent storage device like SSD, HDD.

(iii) Virtual memory:-

- It is an extension of RAM, using disk space to handle large application.



History of computers.

1) First Generation computer (1940-1950)

- First generation computer use vacuum tube for processing and memory which made them large, slow and expensive. ENIAC (Electronic numerical integrator and computer) first computer discovered in 1945. Also UNIVAC discovered in 1951.

Key features:-

- used vacuum tube
- Required large space and high power.

2) Second Generation computer (1950-1960)

- Second Generation computer replace vacuum tube with transistors. Model of computer launched was IBM (1401)

Key features:-

- used transistors instead of vacuum tube.
- Faster and more reliable.
- Store data on magnetic tape and disk.

3) Third Generation computers (1960-1970)

- Third Generation computer used integrated circuit (IC) which combine multiple transistors on a single chip. Transistor controls the flow of electricity. Example: (IBM/360) launched in 1964

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4) Fourth Generation computers (1970-1980)

- Fourth Generation computer use microprocessor where the entire CPU was integrated onto a single silicon chip. IBM PC (1981), Macintosh

Key Features?

- used microprocessor including GUI (Graphical user interface).
- make computer small, easy to use.
- Silicon is more cheaper and good conductor of electricity.

5) Fifth Generation computer (1990-continue)

- Fifth Generation used AI, cloud computing and quantum computing. used multi-core processors, featuring AI and ML capabilities. which helps in performance increment. Example: intel pentium and AI-Super computer.

Logic Gate:

- Logic gate is an electronic circuit that performs a Boolean function.

① And gate:

• Truth table

A	B	C
0	0	0
0	1	0
1	0	0
1	1	1

Boolean expression

$$C = A \cdot B$$

② OR gate

• Truth table

A	B	C
0	0	0
0	1	1
1	0	1
1	1	1

Boolean expression

$$C = A + B$$

③ NOT gate

• Truth table

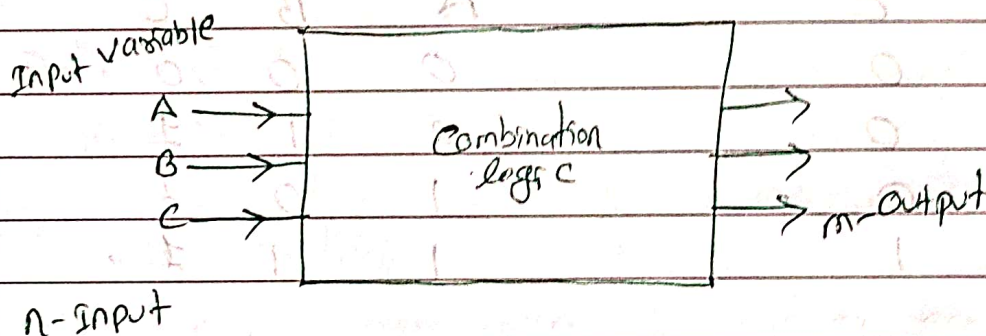
A	C
0	1
1	0

Boolean expression $C = \bar{A}$

Combinational / logical circuit.

- It is a combinational logic like a simple machine that takes input and gives you an output.

Diagram:



How to Design a combinational circuit

- i) write down the problem.
- ii) make a Truth table
- iii) Simplify the boolean algebra.
- iv) Draw the circuit

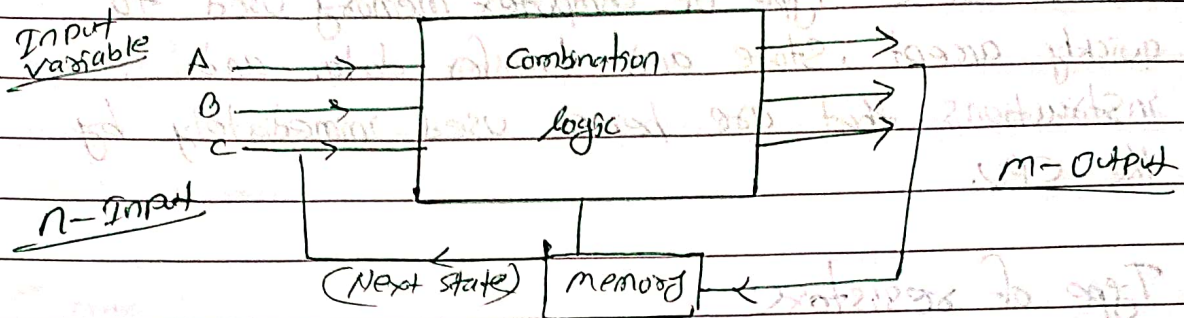
Sequential circuit

- It is like a memory machine, it does not just depend on what you are doing right now. It also remembers what happens before.

How to design a sequential circuit

- i) understand the problem
- ii) create a state diagram
- iii) make a state table
- iv) Choose memory element
- v) write equation and draw a circuit

Diagram:-



Difference between combinational and sequential circuit

Combinational

Sequential

i) The circuit whose output depends only on the current input. The circuit whose output depends on both current and past input.

ii) No memory elements, it does not store past input data. Contains memory elements like flip-flop.

iii) Examples: Adders, multiplexers, decoders. Examples: counter, flip-flop.

(Cache)

Computer registers:-

Registers are a type of computer memory used to quickly accept, store and transfer data and instructions that are being used immediately by the CPU.

Type of registers:-

- 1) Data registers:- (16 bits)
 - Hold data that being transfer to memory.
- 2) Address registers:- (12 bits)
 - Hold the memory address for reading and writing data.
- 3) Status registers:- (8, 16, 32) bits
 - Hold flags or indicators about the state of the CPU.
- 4) Program Counter (PC) (12 bits)
 - keeps track of the next instruction to execute.
- 5) Accumulator (16 bits)
 - A register used for Arithmetic and logical operation.

Instructions, timing and control

Instructions:-

- It is a small computer command that tell a computer what to do.

Timing

- The CPU works ~~at~~ in the ^{step} ~~state~~ control a clock.

Control unit.

- It is act like a manager making sure each step is followed properly.

Instruction cycle:-

1) Fetch

- Get the instruction from memory.

2) Decode

- understand what the instruction means.

3) Execute.

- Perform the instructions.

4) Store

- Save the result.

Memory reference instruction:

- Memory reference instruction are the command in a computer that access memory to either Read or write data.

Example

store in memory

If we want to add two numbers the process will be

- read the first number from memory
- read the second number from memory
- Add both numbers
- store the result back into memory

Common memory reference instruction:

- LDA memory
→ load data from ↓ into the accumulator.
- STA
→ Store data from accumulator into memory
- Add
→ Add data from memory to accumulator.
- Sub
→ Subtract data from memory
- Mul
→ Multiply data

Note: (Accumulator) : It is a special type of register that store the result of arithmetic and logical operation.

Pipelining review:

- Pipelining is a technique used to process to execute multiple instructions at a same time by dividing the execution process into different stages.

Stages of pipeline

i) Instruction Fetch (IF)

→ Get the instruction from memory

ii) Instruction Decode (ID)

→ understand what is the instruction does

iii) Execute:

→ Perform calculation or logical operation.

iv) Memory access.

→ Read and write data from memory.

v) Write back

→ store the result back into the register.

Control Hazard: ▽

- A control hazard happen when the processor doesn't know which instruction to fetch next due to a branch

Example: if-else, loop, function call.

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if (x > 0) then
    y = 5
else
    y = 10
  
```


Comparison table

	Memory type	Speed	Size	Location
i)	Register	Fastest	Small	Inside CPU
ii)	Cache (L1, L2, L3)	Very fast	Small	Close to CPU
iii)	RAM	Slower	Large	External to CPU
iv)	Secondary memory	Slowest	Very large	External